

Quite a few smartphones these days come with optical image stabilization, that stabilizes the lens from physical movements. The lens system (and in some cases the entire camera system) is suspended in an array of springs. The camera takes the input from the accelerometer and gyroscope and determines the camera shake and then moves the camera system in the exact opposite direction of the shake, thereby canceling most of it. OIS cannot completely cancel out all the camera shake but it does help reduce it significantly. This allows you to get smoother videos and better low light images, as the camera can now use a slightly longer shutter speed to allow more light in without having to worry much about camera shake.

The sensor is just as important as the lens. The image sensor is what converts the incoming light into electrical signal, which is then turned into a digital image. The quality and physical size of the image determines the final output of the image. A bigger sensor can naturally capture more light. Two sensors of the same size and lens system will have the same amount of light falling on them. The higher resolution sensor will try to split the same amount of light into more parts, which results in a more detailed image but in low light it also results in darker image as the smaller individual pixels will receive less light than that on a sensor with bigger pixels.

To tackle this situation, HTC recently introduced the UltraPixel camera on their HTC One. Basically, this is like any other camera sensor but instead of putting in more pixels for the same sensor size like everyone else, HTC went the other way by putting in fewer pixels, four to be precise. This results in individual pixels being much bigger than on any other phone, which in theory gives much better results in low light. To HTC's credit, the UltraPixel camera does produce good images in low light but the 4 megapixel resolution is not sufficient to resolve enough detail in daylight conditions.

Advancement in camera technology means we don't have to worry about that too much about pixel size now. We even have advanced tech such as backside-illumination now available on mobile phones, which was once considered too complicated and expensive for these devices. A backside-illuminated sensor places the wiring layer behind the sensor instead of in front of it. This allows more light to fall directly on the sensor instead of first passing through the wiring layer. The sensors themselves are more sensitive now, which allows them to be smaller without output.

The last part of the camera hardware is the flash. Almost all phones that have a flash have an LED or light emitting diode flash. Compared to xenon flash on proper cameras, an LED can stay on for much longer and